

PAM Data Transform Visualizer

AJ, Bri, Ivia?

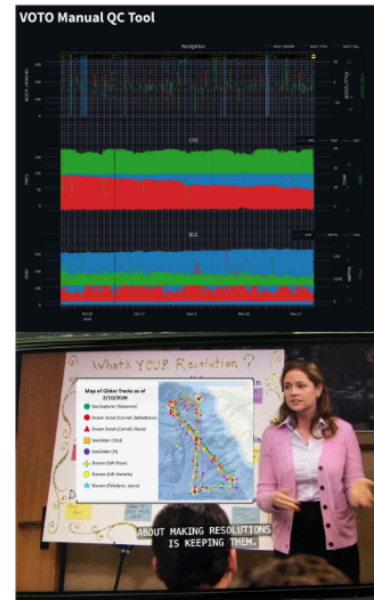


PAM data transform visualizer

- One sentence goal: Build a simple visualizer to see how transformational data operations change PAM data products.

name=pending

- Slack channel: # pam-visualizer
- GitHub repo: /pam-visualizer
- Team members: AJ, <hopefully someone else>
 - have experience with interactive plotting in Python
 - need experience with PAM data (where to find it and how is it structured), operations/common glider processing steps
- Potential Data Sources/Tools
 - Some raw data from authorized sources (NCEI), final data shared
- Potential Challenges
 - Testing on different systems
 - Large file sizes
- First steps?
 - Download the data, try to plot with `plotly`, `bokeh`, or `streamlit`



Here's a [report](#) AJ's worked on, interested in building a tool like the slide on the top

- To be useful to techs, students, etc for downstream processing (or to prepare/learn for downstream processing)
- He has experience interacting experience w/ interactive python development
- Value of documentation—working notebook example as main deliverable

Have side by side versions to see changes in transformations (visual side by side retained)

Helpful for non-PAM people who are now adopting PAM

This could be helpful for mission planning & quality control post-mission (may have different designs).

Has value as a teaching tool and with onboarding students

He's coming from a post-processing perspective, taking residuals on what we gain/lose.

Might be good to have 'best practices' as part of this consideration— maybe references/resources might also be a part of this? +1

-In addition to DOSITs, Bioacoustics Stack Exchange, etc - I recently [found this list of resources to help students understand DSP through lens of music](#), and [also this one](#)

I think Bob Currier at Texas A&M built a handful of tools for event detection and noise reduction.

Typical plot types used to assess PAM data:

Spectrogram

Waveform

Frequency Spectrum

Generated outputs in pre-processing:

- Long-term spectral averages (LTSA) generated and viewable in Triton (MATLAB). These spectrogram visualizations already are products of the pre-processing step so do we try and integrate them directly into this tool so we are not “re-inventing” the wheel? It wouldn't allow you to fine tune since they are generated with pre-set settings.

Data Source?

I think first we need to decide on where our acoustic data is coming from? Is it possible to request short clips of audio from each glider deployment as a test for the visualizer? Not sure where to look on NCEI for glider data to work with. Ivia - do you have glider audio data we can work with to use as examples?

Useful functionality to input marine mammal detection information so folks new to PAM have an idea of what is in the deployment data? I know this deviates from the purpose of the transformer visualizer but maybe an additional add-on? Thinking of how students might use this tool as well.

AJ - I copy and pasted your idea below

Potential expansion *during (live brainstorming) Selene's presentation:

- Once we have a few settings/transformers sorted out that we'd like to analyze, do we notice that the settings do better or poorer on the different phases? I.e., do the FFT settings need to be changed when we're in the FAST mode vs the SLOW or DRIFT modes?
- Similarly, we might be able to do comparisons by using said settings on different models. Is the SeaExplorer noisier than the Slocum (they use different sensors and are different shapes)?